



Day-19

1. Introduction to CRUD Operations

CRUD stands for:

- **C** → **Create**
- **R** → **Read**
- **U** → **Update**
- **D** → **Delete**

These are the **four basic operations** performed on any database.

👉 Every real-world application (Instagram, banking apps, student systems) uses CRUD.

2. CREATE Operation (Adding Data)

◆ Definition

The **CREATE operation** is used to **insert new records** into a database.

Purpose

- Add new data
- Expand database records
- Store user information

♦ Example (Python + MySQL)

```
query="INSERT INTO students (id, name, age) VALUES (%s, %s, %s)"
values= (1,"Mahak",20)

cursor.execute(query,values)
conn.commit()
```

♦ Understanding Placeholders

- `%s` is a **placeholder**
- It prevents **SQL injection**
- Makes code safe and dynamic

👉 Example:

```
VALUES (%s, %s, %s)
```

3. READ Operation (Fetching Data)

♦ Definition

The **READ operation** is used to **retrieve data** from the database.

Purpose

- Display stored data
- Search records
- Show information to users

♦ Example

```
query="SELECT * FROM students"
cursor.execute(query)
```

```
result=cursor.fetchall()
```

```
for row in result:
```

```
    print(row)
```

◆ Key Points

- `fetchall()` → gets all records
- `fetchone()` → gets one record

👉 This is how apps show data on screen.

📌 4. UPDATE Operation (Modify Data)

◆ Definition

The **UPDATE operation** is used to **change existing records**.

🎯 Purpose

- Correct data
- Update user details
- Keep database accurate

◆ Example

```
query="UPDATE students SET name=%s WHERE id=%s"
values= ("Rahul",1)
```

```
cursor.execute(query,values)
conn.commit()
```

◆ Key Points

- Always use `WHERE` condition ⚠️
- Without it → all records may change

5. DELETE Operation (Remove Data)

◆ Definition

The **DELETE operation** removes data from the database.

Purpose

- Remove unwanted records
- Clean database
- Maintain relevant data

◆ Example

```
query="DELETE FROM students WHERE id=%s"  
value= (1,)
```

```
cursor.execute(query,value)  
conn.commit()
```

◆ Warning

```
DELETEFROM students;
```

👉 This deletes **ALL records** (dangerous)

6. Student Management System (Real-Life Example)

A **Student Management System (SMS)** uses CRUD operations.

◆ Features

Operation	Function
CREATE	Add student
READ	View students
UPDATE	Edit details

Operation	Function
DELETE	Remove student

♦ Example Table

```
CREATETABLE students (  
    idINTPRIMARYKEY,  
    nameVARCHAR(50),  
    ageINT,  
    courseVARCHAR(50)  
);
```

7. Key Takeaways

- CRUD is the **foundation of backend development**
- Every database system uses CRUD
- Python + MySQL makes data handling easy
- Understanding CRUD = Building real applications

8. Final Understanding

If a student understands:

- Insert (Create)
- Select (Read)
- Update
- Delete

👉 Then they are already thinking like a **backend developer**